

# Fernando Fernández Andueza

Gameplay & AI Programmer

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Valencia, Spain · EU citizen · open to relocation within Spain and to remote work · Spanish (native), English (professional)

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## PROFILE

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Gameplay and AI programmer working in Unreal Engine 5 (C++ and Blueprints). Shipped Beerserker: The Stolen Brew on Steam, owning the enemy AI in StateTree. Wrote XEMA, a custom C++ engine, from scratch — OpenGL 4.5, deferred PBR rendering, an ECS core and a Nintendo Switch port, with a PlayStation 4 port in progress. Two prior years as a fullstack developer shipping ERP software in production.

## EXPERIENCE

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**Gameplay & AI Programmer** · Beerserker: The Stolen Brew (<https://store.steampowered.com/app/4522440/>) · ESAT final-year production, published on Steam by Grumpy Games

Sept 2025 – Jul 2026

Enemy AI on a 10-month production built by a 20+ person multidisciplinary team at ESAT and released commercially on Steam. Designed and implemented four distinct enemy archetypes in StateTree (Unreal Engine 5, C++) — melee, ranged and explosive enemies, plus a tank mini-boss with four unique attack patterns.

**Unity Game Developer** · Roadtrip: The Engine of Madness ([https://store.steampowered.com/app/3776540/Roadtrip\\_EL\\_Motor\\_de\\_la\\_Locura/](https://store.steampowered.com/app/3776540/Roadtrip_EL_Motor_de_la_Locura/)) · Oniric Tales

Dec 2024 – Jun 2025

Design, architecture and development of core gameplay systems. Built a scalable component-based dialogue system, an AI perception system integrated into behaviour trees, and a radio communication system; contributed to the Behavior Graph, general gameplay mechanics, profiling and bug fixing.

**Fullstack Developer (ERP)** · Enterprise software industry

Sept 2020 – Feb 2023

Built scalable web applications and enterprise services with Vue.js, TypeScript and Microsoft SQL Server. Managed production server environments and resolved live client incidents under strict SLAs, working in Agile teams as a certified Scrum Product Owner and Scrum Master.

## SELECTED PROJECTS

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### XEMA — custom 3D engine (<https://andueza013.github.io/PortfolioGameDev/graphic-engine>)

C++ · OpenGL 4.5 · Nintendo Switch

Solo-built engine, 10 months. ECS core storing components in contiguous arrays with O(1) access, deferred renderer with PBR, a std::future job system driving asynchronous asset loading, multithreaded Wave Function Collapse for biome generation, Lua scripting through sol2 and PhysX 5 integration. Ported to Nintendo Switch via libnx and EGL; a PlayStation 4 port to GNM/GNMX is in progress.

### Abandoned Hospital

Unreal Engine 5 · C++ · AI Perception, EQS

Enemy and NPC behaviour driven by AI Perception and EQS in Unreal Engine. First implemented with behaviour trees, then rebuilt from scratch on StateTree to compare both approaches to the same problem.

## EDUCATION & CERTIFICATIONS

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### HND in Video Game Programming · ESAT — Escuela Superior de Arte y Tecnología · Valencia 2023 – 2025

Specialisation in low-level systems and game loops. Built XEMA, a cross-platform 3D engine from scratch in C++ (OpenGL 4.5 DSA, deferred shading, PBR), implemented ECS frameworks with O(1) access constraints, and ported the codebase to Nintendo Switch.

### DAM — Multiplatform Application Development · IES El Grao · Valencia 2018 – 2020

Object-oriented programming, relational databases (SQL) and application design patterns; mobile and desktop development, client-server architecture and version control workflows.

### Scrum Master & Product Owner · Scrum.org / International Agile Institutes Professional credentials

Certified in both roles: bridging design requirements and system architecture, optimising production pipelines and managing technical debt across team iterations.

## TECHNICAL SKILLS

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|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Game engines</b>              | Unreal Engine 5 (C++ & Blueprints), Unity (C#), XEMA — custom C++ engine                                                                                 |
| <b>AI &amp; gameplay</b>         | StateTree, Behaviour Trees, EQS, AI Perception, Animation systems, Object pooling                                                                        |
| <b>Languages</b>                 | C++, C#, C, Lua                                                                                                                                          |
| <b>Graphics &amp; systems</b>    | OpenGL 4.5 DSA, Deferred shading / PBR, Vertex & fragment shaders, ECS architecture, Multithreading & job systems, Memory management & pointers, PhysX 5 |
| <b>Debugging &amp; profiling</b> | Unreal Insights, Visual Studio debugger, RenderDoc                                                                                                       |
| <b>Platforms</b>                 | PC (Windows), Nintendo Switch, PlayStation 4 (in progress)                                                                                               |
| <b>Tooling</b>                   | Git, Perforce, Plastic SCM, Premake5                                                                                                                     |
| <b>Outside games</b>             | TypeScript / JavaScript, Vue.js, SQL Server, Java, VB .NET                                                                                               |

Last updated: September 2026